

Static

Transistor Count 32	A<B	A=B	A>B
Rise Time	636.94268ps	636.94268ps	477.70701ps
Fall Time	318.47134ps	159.23567ps	263.57827ps
Propagation Delay	477.705ps	398.0875ps	370.64ps
Avrage Power	130.61pW		
Product Delay Propagation	0.000062f	0.000051f	0.000048f

TG

Transistor Count 16	A<B	A=B	A>B
Rise Time	838.92617ps	503.3557ps	335.57047ps
Fall Time	503.3557ps	671.14094ps	671.14094ps
Propagation Delay	671.11ps	587.24ps	503.35ps
Avrage Power	105.21uW		
Product Delay Propagation	70.6f	61.78f	52.95f

PASS

Transistor Count 10	A<B	A=B	A>B
Rise Time	621.11801ps	590.4908ps	759.20245ps
Fall Time	310.55901ps	253.06748ps	253.06748ps
Propagation Delay	465.83ps	421.77ps	516.03ps
Avrage Power	38.12pw		
Product Delay Propagation	0.000017f	0.000016f	0.000019f

GDI My Propose

Transistor Count 10	A<B	A=B	A>B
Rise Time	553.02804ps	515.35826ps	532.37224ps
Fall Time	154.20561ps	157.72871ps	158.18612ps
Propagation Delay	353.5ps	336ps	345ps
Avrage Power	34.18pw		
Product Delay Propagation	0.0000120f	0.0000114f	0.0000117f

2025 paper propose(PTL-GDI-TG)			
Transistor Count 24	A<B	A=B	A>B
Rise Time	490.19608ps	816.99346ps	653.59477ps
Fall Time	980.39216ps	163.39869ps	163.39869ps
Propagation Delay	730.29ps	490.019ps	408.49ps
Avrage Power	80.80 uw		
Product Delay Propagation	59f	39.59f	33f

CMOS Tacnique	Propagation Delay	Avrage Power Dicipiation	Product Delay Propagation
Improved Design of CMOS 1-Bit Comparator With Stacking Technique (2017)			
Static CMOS	49.38 ns	174.352 pW	0.0086 f
Design of CMOS Based 1-bit Comparator with MTCMOS and Forced Stack Technique in 180 m Technology (2019)			
Static CMOS	330.61 ps	48.12 pW	0.0000159 f
Design_of_a_Two-Bit_Magnitude_Comparator_Based_on_Pass_Transistor_Transmission_Gate_and_Conventional_Static_CMOS_Logic(2020)			
Static, PASS,TG	0.192 ns	7.792 μ W	1.496f
Design and Analysis of a 1-bit Comparator for Low Power VLSI Systems(2025)			
6T MUX ADDER 1 BIT Comaprator	5.5 ps	7.3 μ W	0.04 f
MY Popose			
GDI	336ps	34.18pw	0.0000114f